

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 1: Radio Wave Propagation, Antennas & Frequencies

Syllabus Source: CAE Oxford (pp. 1-44) | Pass Target: >= 90% AviationExam

1. THE ELECTROMAGNETIC SPECTRUM & FREQUENCY BANDS

Radio waves travel at the speed of light ($c = 300,000 \text{ km/s} = 3 \times 10^8 \text{ m/s}$). Wavelength is inversely proportional to frequency: $\lambda \text{ (meters)} = c / \text{frequency (Hz)}$:

Frequency Band	Frequency Range & Wavelength	Aviation Navigational Aids & Communications
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VLF (Very Low)	3 to 30 kHz (100 to 10 km)	Submarine communications, Omega navigation (obsolete).
LF (Low Frequency)	30 to 300 kHz (10 to 1 km)	NDB (Non-Directional Beacons), commercial AM radio.
MF (Medium Frequency)	300 to 3,000 kHz (1 km to 100 m)	NDB, Locator beacons, maritime radio.
HF (High Frequency)	3 to 30 MHz (100 to 10 m)	Long-range transoceanic voice communications (Sky wave).
VHF (Very High)	30 to 300 MHz (10 to 1 m)	VHF Comms (118-137 MHz), VOR (108-118 MHz), ILS Localizer (108-112 MHz).
UHF (Ultra High)	300 to 3,000 MHz (1 m to 10 cm)	ILS Glide Path (329-335 MHz), DME (960-1215 MHz), SSR (1030/1090 MHz), GPS L1 (1575 MHz).
SHF (Super High)	3 to 30 GHz (10 to 1 cm)	Airborne Weather Radar (AWR 9.3 GHz), MLS (5 GHz), Radar Altimeter (4.3 GHz).

2. WAVE PROPAGATION MODES & MAXIMUM RANGE

- Ground Wave (Surface Wave): Follows the curvature of the Earth. Dominant in LF and MF bands (NDB). Range is greatest over seawater (high conductivity) and lowest over dry sand/ice.
- Sky Wave (Ionospheric Reflection): Reflected by ionosphere layers (D, E, F1, F2). Dominant in HF band. Subject to skip distance, skip zones, and severe night effect fading.
- Space Wave (Line-of-Sight): Straight-line propagation in VHF, UHF, and SHF bands. Limited by atmospheric refraction and Earth curvature: Maximum Theoretical Range (NM) = $1.23 \times (\sqrt{h_{\text{transmitter}}} + \sqrt{h_{\text{receiver}}})$ [heights in feet]!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: VHF Maximum Line-of-Sight Radio Range

SCENARIO (Fundamental AviationExam Question): An aircraft is cruising at FL 360 (36,000 ft). A VOR ground station antenna is located at an elevation of 900 ft MSL. QUESTION: What is the maximum theoretical range in Nautical Miles at which the aircraft can receive the VOR signal? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Transmitter and Receiver Heights in feet: - Aircraft Height (h_{rx}) = 36,000 ft - VOR Ground Antenna Height (h_{tx}) = 900 ft Step 2: Recall the Master VHF Line-of-Sight Range Formula: - Formula: Range (NM) = $1.23 \times [\sqrt{h_{tx}} + \sqrt{h_{rx}}]$ Step 3: Calculate the square roots of each height: - $\sqrt{900} = 30$ - $\sqrt{36,000} = 189.74$ - Sum of square roots = $30 + 189.74 = 219.74$. Step 4: Multiply by the atmospheric refraction constant (1.23): - Range (NM) = $1.23 \times 219.74 = 270.28$ NM (~270 Nautical Miles)! FINAL ANSWER: The maximum theoretical VOR reception range is approximately 270 Nautical Miles.

[!] EXAM TRAP (AVIATIONEXAM) - The 1.05 vs 1.23 Range Constant Trap

- If both antennas are optical line-of-sight in a vacuum: constant = 1.05. - In the Earth's atmosphere, radio waves BEND SLIGHTLY downwards due to atmospheric refraction: constant = 1.23 (or 1.25)! - EASA exams universally use 1.23: Range = $1.23 \times (\sqrt{h_1} + \sqrt{h_2})$.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: In which frequency band does the Instrument Landing System (ILS) LOCALIZER operate?**

[A] UHF [B] VHF (108.10 MHz to 111.95 MHz, odd tenths) [C] SHF [D] LF

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The ILS Localizer operates in the VHF band between 108.10 and 111.95 MHz on odd-tenth frequencies (e.g. 109.10, 109.30). The Glide Path operates in UHF.

Q2: Over which surface type does an LF/MF Ground Wave travel with the GREATEST range and minimum signal loss?

[A] Dry desert sand [B] SEAWATER (due to high electrical conductivity) [C] Dense forest [D] Mountainous rocky terrain

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Seawater has the highest electrical conductivity of any natural Earth surface, producing minimum ground attenuation and maximum surface wave range.

Q3: What is the wavelength of a radar operating on a frequency of 10 GHz (10,000 MHz)?

[A] 3 meters [B] 3 CENTIMETERS ($\lambda = 300,000,000 \text{ m/s} / 10,000,000,000 \text{ Hz} = 0.03 \text{ m} = 3 \text{ cm}$) [C] 30 cm [D] 30 meters

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Formula: $\lambda = c / f$. For 10 GHz: $3 \times 10^8 / 10 \times 10^9 = 0.03 \text{ m} = 3 \text{ cm}$ (standard Airborne Weather Radar X-band).